

TopoLiDM: Topology-Aware LiDAR Diffusion Models for Interpretable and Realistic LiDAR Point Cloud Generation

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Abstract—LiDAR scene generation is critical for mitigating real-world LiDAR data collection costs and enhancing the robustness of downstream perception tasks in autonomous driving. However, existing methods commonly struggle to capture geometric realism and global topological consistency. Recent LiDAR Diffusion Models (LiDMs) predominantly embed LiDAR points into the latent space for improved generation efficiency, which limits their interpretable ability to model detailed geometric structures and preserve global topological consistency. To address these challenges, we propose TopoLiDM, a novel framework that integrates graph neural networks (GNNs) with diffusion models under topological regularization for high-fidelity LiDAR generation. Our approach first trains a topological-preserving VAE to extract latent graph representations by graph construction and multiple graph convolutional layers. Then we freeze the VAE and generate novel latent topological graphs through the latent diffusion models. We also introduce 0-dimensional persistent homology (PH) constraints, ensuring the generated LiDAR scenes adhere to real-world global topological structures. Extensive experiments on the KITTI-360 dataset demonstrate TopoLiDM’s superiority over state-of-the-art methods, achieving improvements of 22.6% lower Fréchet Range Image Distance (FRID) and 9.2% lower Minimum Matching Distance (MMD). Notably, our model also enables fast generation speed with an average inference time of 1.68 samples/s, showcasing its scalability for real-world applications. We will release the related codes at <https://github.com/IRMVLab/TopoLiDM>.

I. INTRODUCTION

Recently, LiDAR data has achieved remarkable research focus with the advantages of accurate geometric measurements [1]–[3], facilitating various downstream tasks in autonomous driving [4]–[6], robotic localization [7], [8], and SLAM system [9]–[11]. However, the LiDAR data collection process remains difficult due to the substantial costs of physical LiDAR sensors and performance degradation in corner cases, e.g., extreme weather conditions [12]. Therefore, many researchers resort to deep generative models for realistic and low-cost LiDAR scene generation [13]–[17].

However, existing methods for LiDAR data generation often face significant challenges in maintaining geometric fidelity, topological consistency, and interpretable ability.

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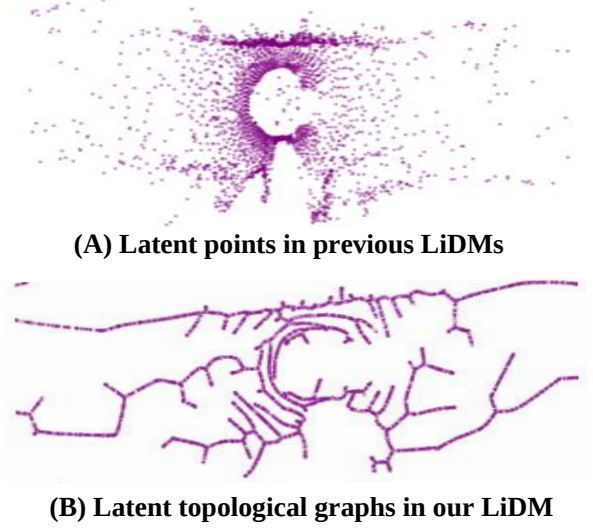


Fig. 1: Comparison with previous LiDAR Diffusion Models (LiDMs). Previous LiDMs commonly adopt latent points as the variables in diffusion models, which fail to consider the global topological consistency. Our method treats topologically-regularized graphs as latent variables to generate interpretable LiDAR graphs with geometric correlations.

Caccia et al. [13] propose an unsupervised method with Generative Adversarial Networks (GAN) [18] and Variational Autoencoders (VAEs) [19]. Nonetheless, the training process of GAN suffers from instability and mode collapse, and the generation samples commonly lack geometric details [17]. More recently, many researchers also explore LiDAR generation based on diffusion models [14], [16], [17]. These methods predominantly embed LiDAR points into the latent space to improve the generation efficiency as in Fig. 1 (A), but this limits the interpretable ability to model geometric correlations and preserve global topological consistency.

To address these challenges, we propose TopoLiDM, a novel framework that integrates graph neural networks (GNNs) with diffusion models under topological regularization for interpretable and high-fidelity LiDAR point generation. TopoLiDM consists of a two-stage training paradigm. First, a topology-preserving VAE learns how to compress input LiDAR range images into compact topological graphs and integrates 0-dimensional persistent homology (PH) constraints [20] into the learning objective, which ensures that synthesized LiDAR scenes maintain the intrinsic global topological properties inherent to real-world distributions.

The VAE encoder incorporates a computational graph transformation module that converts input points into graph-structured representations, followed by a series of GNN layers to establish topological relationships. Then, a diffusion-based generation module is designed based on compact and interpretable topological graphs as the intermediate latent variables, effectively preserving geometric correlations. This compact representation not only optimizes the generation efficiency but also enhances the model’s capacity to capture long-range dependencies and enforce consistency in global topological structures. Finally, generated topological graphs are decoded into 3D LiDAR points by the pre-trained topology-aware decoder of VAE.

Overall, our key contributions are as follows:

- We propose TopoLiDM, a novel framework integrating GNNs with diffusion models under topological regularization. With compact topological graphs as latent representations, our method can realize fast, interpretable, and high-fidelity LiDAR generation.
- We design a topology-aware VAE module that captures long-range dependencies and enforces 0-dimensional persistent homology constraints, ensuring the generated LiDAR scenes adhere to the global topology rules of real-world environments.
- Extensive experiments on the KITTI-360 dataset demonstrate that TopoLiDM achieves significant improvements over state-of-the-art methods. Notably, our model surpasses baseline methods by 22.6% in Fréchet Range Image Distance (FRID) and 9.2% in Minimum Matching Distance (MMD) with fast inference time, showcasing great potential for practical applications.

II. RELATED WORK

A. Deep Generative Models

Deep generative models, including GAN [18], VAE [19], etc., have revolutionized the field of computer vision and machine learning, enabling the synthesis of high-fidelity 2D images [21] and 3D points [22]. Recently, diffusion models [23] have emerged as a more advanced generative backbone with better generation quality and diversity, leveraging a progressive denoising process to generate realistic samples. Latent diffusion models (LDMs) [21] further improve computational efficiency by operating in a perceptually equivalent latent space. Diffusion models have been successfully applied to various tasks, including text-to-image synthesis [24], image-to-image translation [25], and 3D shape generation [22]. However, the exploration of diffusion models in LiDAR data generation remains limited, primarily due to intrinsic challenges from sparse and irregular LiDAR points.

B. LiDAR Data Generation

LiDAR data generation is a critical task for robot localization [26]–[28] and autonomous driving [29], [30] because it enables the creation of training data to enhance the performance in downstream perception tasks [31]. Traditional methods for LiDAR data generation often rely on physics-based simulators or data-driven approaches [32]. While these

methods can produce realistic LiDAR scenes, they often fail to capture the intricate details and global topological properties of real-world environments. Recent works have explored the use of deep generative models for LiDAR data generation, including variants of GANs [13], VAEs [15], and diffusion models [16], [17]. However, these methods often struggle to maintain geometric fidelity, topological consistency, and interpretable ability in latent space.

C. Topological Learning for 3D Points

Topological machine learning [33] is an emerging field that combines topological data analysis with machine learning techniques to capture and quantify the global shape and evolution of topological features in datasets. Persistent homology (PH) [34] is a key tool in this field, enabling the extraction of topological invariants such as connected components, loops, and voids. Recent works including [34]–[36] have explored the integration of PH with deep learning models, demonstrating its potential for improving the robustness and generalization of neural networks. In the context of LiDAR data generation, topological regularization has been shown to enhance the realism and fidelity of generated scenes by enforcing global topological constraints [20], [37]. This motivates us to design topological regularization and PH-based constraints in diffusion models to ensure the generated LiDAR scenes adhere to the intrinsic global topology of real-world environments.

III. METHOD

A. Overall Architecture

As in Fig. 2, given an input LiDAR range image $I \in \mathbb{R}^{H \times W \times C}$, TopoLiDM first encodes it into a compact graph latent feature $Z = \mathcal{E}(I) \in \mathbb{R}^{h \times w \times c}$ using a graph encoder $\mathcal{E}$. The encoded graph Z consists of nodes V representing local point clusters and edges E that encode spatial relationships through hierarchical graph convolutions.

In the diffusion stage, a UNet [38] backbone $\mathcal{E}_\theta$ learns to denoise latent representations Z_t over T timesteps, guided by diverse input conditions, e.g., text, trajectory, etc. The diffusion process operates entirely in the latent space, enabling efficient LiDAR generation.

Finally, the decoder $\mathcal{D}$ reconstructs the range image $\hat{I} = \mathcal{D}(Z)$ utilizing a combination of upsampling modules and circular convolutional blocks [17].

B. Topology-Preserving Autoencoding

To achieve efficient generation on latent space and preserve the accurate topology relationships, we first employ a topology-preserving VAE inspired by [20] to compress the input LiDAR range image $I \in \mathbb{R}^{H \times W \times C}$ into a compact latent representation $\mathbf{z}_0 \in \mathbb{R}^{h \times w \times c}$, where the downsampling factors $f_v = \frac{H}{h}$, $f_h = \frac{W}{w}$ determine the vertical and horizontal compression ratio, respectively. This latent representation $\mathbf{z}_0$ captures the essential topological features by constructing hierarchical graph layers while significantly reducing computational costs.

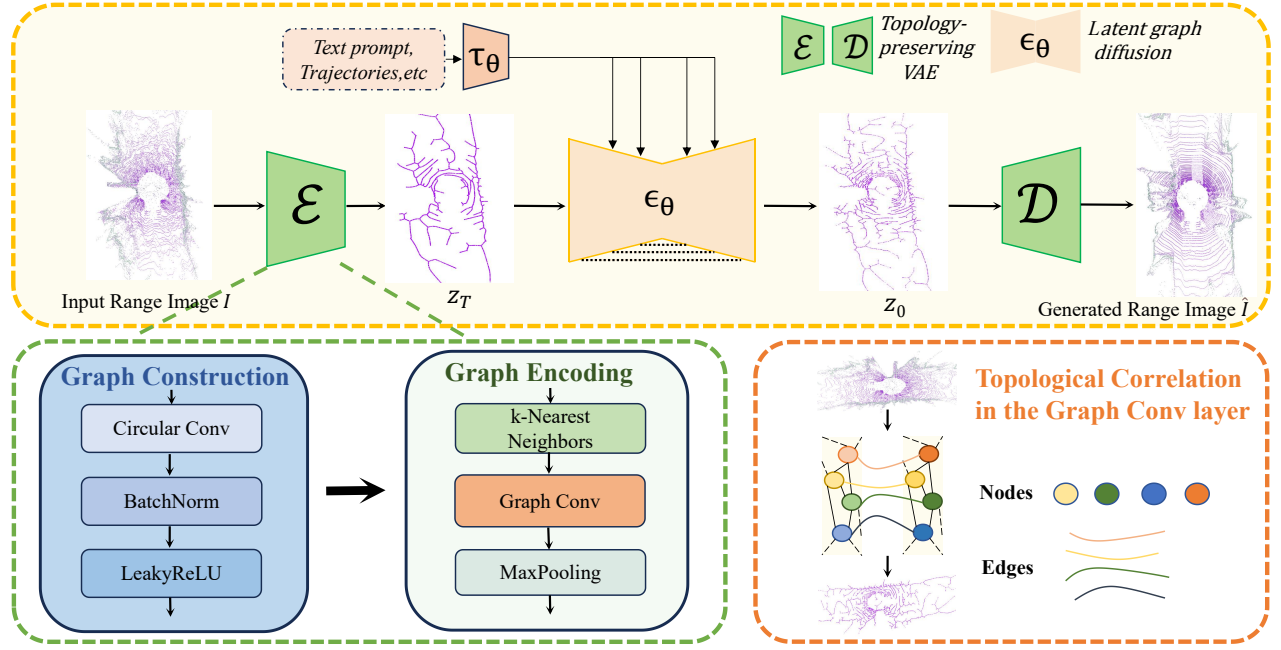


Fig. 2: The overall architecture of TopoLiDM consists of two cascaded stages: (1) **Topology-Preserving Autoencoding** that compresses LiDAR points into graph-structured latent representations. (2) **Diffusion-based Topological LiDAR Generation** that leverages the low-dimensional, semantically-rich representation for a more efficient and interpretable denoising process.

Graph Construction: Following [39], we divide an input range image I into N patches, then apply circular convolution layers to embed each patch into a feature vector $x_i \in \mathbb{R}^D$, where D is the feature dimension and $i = 1, 2, \dots, N$. These features can be viewed as a set of unordered nodes $X = [x_1, x_2, \dots, x_N]$. We also add a positional encoding vector to each node in order to represent the positional information:

$$x_i \leftarrow x_i + e_i,$$

where $e_i \in \mathbb{R}^D$. Edges are established by correlating each node to its k -nearest neighbors.

Hierarchical Graph Encoding: The encoder $\mathcal{E}$ then employs multiple graph layers for effective graph encoding. Specifically, each layer employs message passing to gather information from both immediate and distant neighbors, followed by a max-pooling operation to update the node embeddings as [20]:

$$h_v^{(l)} = \max_{u \in N_v} \left\{ \text{LeakyReLU} \left(W_{\text{conv}}^{(l)} \cdot \sum_{u \in N_v} h_u^{(l-1)} \right) \right\},$$

where N_v is k nearest neighbors for each node v , $W_{\text{conv}}^{(l)}$ is the learnable weight matrix for layer l , and $h_u^{(l)}$ is the embedding of node u at layer l .

This process effectively projects computational graph's node features into higher-dimensional feature $\mathbf{z}_0$, thereby enriching their representation capacity for high-level semantic information. Upon completion of each layer, the original graph structure is discarded, and the updated node embeddings serve as the input for the subsequent layer.

This recursive transformation enables the aggregation of long-range topological relationships, progressively encoding global topological features as in Fig. 3. The encoder is constructed by stacking L LiDAR graph layers, progressively enhancing the global topological features of the input data.

LiDAR Decoding: Then the encoded graph representation (as in the last column in Fig. 3) will serve as the variable in diffusion models (Section III-C) for the iterative LiDAR denoising and generation. After obtaining generated latent graphs, we leverage a circular-convolutional decoder $\mathcal{D}$ [17] to recover LiDAR range images from the latent graph space.

Persistent Homology Constraints: To enforce the alignment between generated LiDAR graphs and the real-world LiDAR points, we enforce a 0-dimensional PH constraints on the decoded range image $\hat{I}$ and penalize deviations from expected topological structures as in Section III-D.

C. Diffusion-based Topological LiDAR Generation

In the second stage, we freeze the pre-trained VAE and utilize a latent diffusion model (LDM) [21] to efficiently generate latent topological features $\mathbf{z}_0$.

During the forward diffusion process, noise is progressively added to the ground truth latent variable $\mathbf{z}_0$ over a series of time steps $t \in \{1, 2, \dots, T\}$, resulting in a noisy latent variable $\mathbf{z}_t$. This process can be formally described as:

$$\mathbf{z}_t = \sqrt{1 - \beta_t} \mathbf{z}_{t-1} + \sqrt{\beta_t} \epsilon_t, \quad \epsilon_t \sim N(0, \mathbf{I}),$$

where β_t is a predefined noise schedule that controls the variance of the noise added at each step.

During the backward denoising process, we employ a time-conditional U-Net network $\epsilon_\theta(\mathbf{z}_t, t)$ to predict the noise

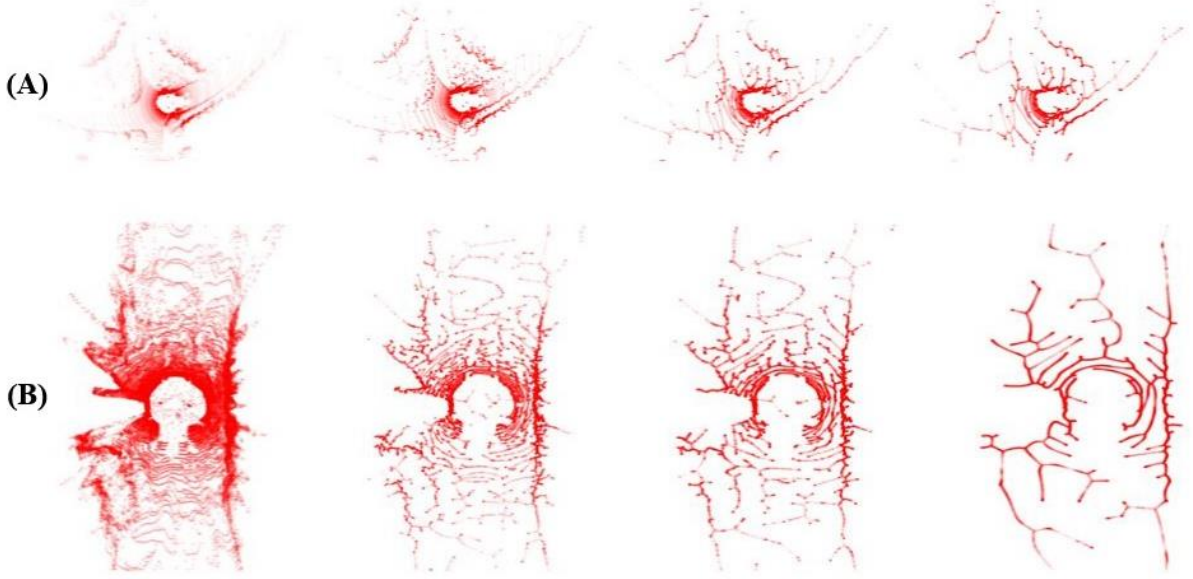


Fig. 3: Visualization of the progressive graph encoding process for latent topological graph creation. From the left to the right, compact graphs are progressively constructed with more topological correlations. The last columns indicate the final latent topological graphs which would be used as the intermediate latent variables in diffusion models.

ε_t added to the noisy latent variable $\mathbf{z}_t$. We leverage the standard loss in diffusion models as introduced in the next section to encourage the accurate prediction of the noise added at each time step, thereby enabling the reverse diffusion process to generate diverse and high-quality latent topological variables $\mathbf{z}_0$ from noisy latent variable $\mathbf{z}_t$. Notably, our LiDAR generation can also be conditioned by additional guidance signals as in Fig. 2, e.g., text prompts, trajectories, which offers more controllability and diversity for conditional LiDAR generation.

D. Training Process and Loss Functions

Our training process consists of two parts: (1) First, we train the topology-preserving VAE, which effectively extracts the latent geometric graph features from the original input LiDAR points and the reverse decoding process. (2) Second, we freeze the parameters of encoder $\mathcal{E}$ and decoder $\mathcal{D}$ in the VAE, and continue to train the latent diffusion models.

Stage 1: VAE Optimization: The loss functions in VAE combine a 0-dimensional PH loss for effective latent topological feature learning, and also a reconstruction loss. The former is a regularization term that leverages 0-dimensional PH to ensure the reconstructed point cloud follows the single connected component, which can be viewed as the global LiDAR topology graphs as in the last column of Fig. 3, thereby preserving the global topological structure of the real LiDAR scan. Formally, the topological loss is defined as:

$$\mathcal{L}_{\text{topo}} = \sum_{i=2}^M \mathbb{1}\{i \neq 1\} (b_i - d_i),$$

where M is the number of connected components, b_i and d_i denote the *birth* and *death* times of the i -th connected

component (0-dimensional topological feature) in the persistence diagram computed via sublevel set filtration over the LiDAR point cloud [20]. The difference $(b_i - d_i)$ quantifies the *persistence* (lifetime) of the component. We also leverage the PH-based loss at the computational graphs from the second and fourth graph layer’s output in the encoder $\mathcal{E}$ to further improve topological embedding. The loss function in the VAE $\mathcal{L}_{\text{VAE}}$ is formulated as:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\|I - \hat{I}\|_1}_{\text{Reconstruction}} + \underbrace{\mathcal{L}_{\text{topo}}(\hat{I}) + \mathcal{L}_{\text{topo}}(L_2) + \mathcal{L}_{\text{topo}}(L_4)}_{\text{Topology}},$$

where L_2 and L_4 represent the updated node embeddings of the computational graphs obtained from the second and fourth layers of the encoder, respectively.

Stage 2: Diffusion Training: During the second training stage, we generate the denoised latent graphs from the latent diffusion model (LDM). To minimize the distribution differences between the ground truth noise and predicted noise, we formulate the standard diffusion loss as:

$$\mathcal{L}_{\text{LDM}} = \mathbb{E}_{\varepsilon \sim N(0, \mathbf{I}), t \sim \text{Uniform}(1, T)} \left[\|\varepsilon - \varepsilon_{\theta}(\mathbf{z}_t, t, \tau_{\theta}(y))\|_2^2 \right],$$

where y is the optional text condition embedded by $\tau_{\theta}(y)$ as in Fig. 2 that guides the diffusion model.

IV. EXPERIMENTS

A. Experimental Setup

Datasets: We evaluate the performance of TopoLiDM on the KITTI-360 dataset [40], which provides high-quality 64-beam LiDAR data for urban driving scenarios, making it ideal for studying distributions of outdoor LiDAR points and also assessing the performance of generated LiDAR scenes in terms of both realism and topological coherence.

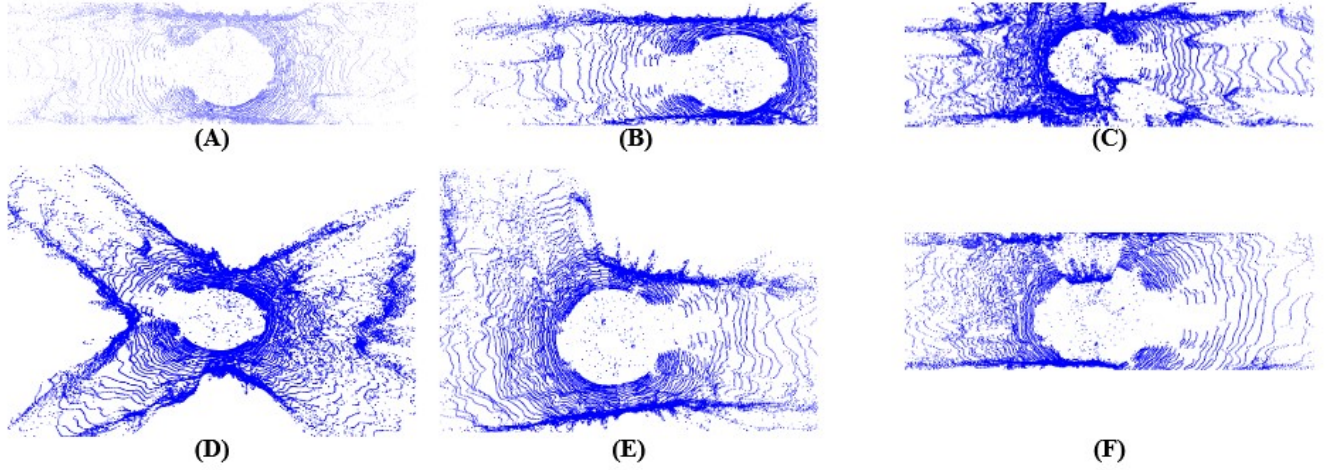


Fig. 4: Visualization of the unconditional generation samples.

TABLE I: Unconditional generation results on the KITTI-360 dataset [40]. Best values are highlighted in **bold**.

Method	Perceptual			Statistical	
	FRID↓	FSVD↓	FPVD↓	JSD↓	MMD↓(10^{-4})
Noise	3277	497.1	336.2	0.360	32.09
LiDARGAN [13]	1222	183.4	168.1	0.272	4.74
LiDARVAE [13]	199.1	129.9	105.8	0.237	7.07
LiDARGen [14] (1160s)	129.1	39.2	33.4	0.188	2.88
LiDARGen [14] (50s)	2051	480.6	400.7	0.506	9.91
LDM(50s) [21]	199.5	70.7	61.9	0.236	5.06
LiDM(50s) [17]	158.8	53.7	42.7	0.213	4.46
TopoLiDM (50s)	118.5	55.9	40.1	0.211	3.86

TABLE II: Efficiency analysis for LiDAR generation.

Method	Speed (samples/s) ↑	Method	Speed (samples/s) ↑
LiDARGen [14]	0.08	LDM [21]	1.31
LiDM [17]	1.74	Ours	1.68

Metrics: We evaluate the performance of TopoLiDM using metrics that assess both the geometric accuracy and the semantic consistency of the generated LiDAR scenes [17]: (1) Fréchet Range Image Distance (FRID): This metric evaluates the perceptual similarity between the generated range images and the real range images. (2) Fréchet Sparse Volume Distance (FSVD): This metric assesses how well the sparse volumes derived from the generated range images match those from the real data. (3) Fréchet Point-based Volume Distance (FPVD): This metric compares the geometric accuracy between point clouds derived from the generated range images and the ground truth point clouds. (4) Jensen-Shannon Divergence (JSD): This is a statistical measure of the distribution similarity between the generated and real LiDAR scenes. (5) Minimum Matching Distance (MMD): This metric evaluates the minimum distance required to match each point in the generated scene with its counterpart in the real scene.

Implementation Details: We evaluate our TopoLiDM on a single NVIDIA RTX 4090 GPU. The model is trained using the Adam optimizer with $\beta_1 = 0.5$, $\beta_2 = 0.9$. The base learning rate is 4.5×10^{-6} for the VAE module, 1.0×10^{-6} for LDM, and both follow a cosine annealing schedule with a maximum period of 100 epochs, dynamically adjusting the

TABLE III: Ablation studies of Topology-aware Encoder.

Method	FRID ↓	FSVD ↓	FPVD ↓	JSD ↓	MMD ↓ (10^{-4})
w/o Topology-aware Encoder [17]	158.8	53.7	42.7	0.213	4.46
with Topology-aware Encoder	118.5	55.9	40.1	0.211	3.86

TABLE IV: Ablation studies of the depth of GNN layers.

Layers	FRID ↓	FSVD ↓	FPVD ↓	JSD ↓	MMD ↓ (10^{-4})
2	140.9	61.7	48.0	0.217	4.31
3	133.2	65.7	44.1	0.219	4.09
4	118.5	55.9	40.1	0.211	3.86

learning rate until it reaches 0. The dimension of input range images is set as $H = 64$, $W = 1024$, $C = 1$. For the Graph Construction module, we set the number of neighboring points as $k = 20$. The number of hierarchical graph layers is set as 4. The downsampling factors f_v is set to 4, f_h is set to 8. The latent dimension in the diffusion model is set to $d = 16$ for a compact representation. The diffusion process is configured with 1,000 timesteps, while the sampling procedure employs 50 steps for efficient generation.

B. Unconditional Generation Results

Quantitative Comparison Results: We compare TopoLiDM with several state-of-the-art methods for unconditional LiDAR data generation, such as LiDM [17]: a recent SOTA approach focusing on generating realistic LiDAR scenes using diffusion models. We also compare other models including LiDARGAN [13], LiDARVAE [13], LiDARGen [14], LDM [21]. As shown in Table I, our TopoLiDM outperforms the chosen baseline method LiDM [17] in most metrics. TopoLiDM also outperforms other state-of-the-art methods in key metrics. It achieves the lowest FRID (118.5), FPVD (40.1), JSD (0.211), and MMD (3.86), indicating superior geometric accuracy and semantic consistency.

Inference Speed: We also evaluate the generation speed in Table II. TopoLiDM demonstrates fast generation speed with an average of 1.68 samples per second, offering potential for real-world applications. Compared with the chosen baseline LiDM [17] (1.74 samples per second), introducing our

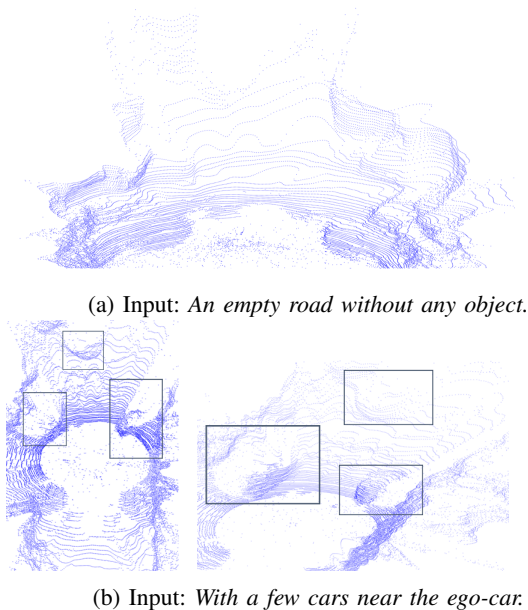


Fig. 5: Visualization samples of the conditional text-to-LiDAR generation.

TABLE V: Ablation studies of Topological Loss.

Method	FRID ↓	FSVD ↓	FPVD ↓	JSD ↓	MMD ↓ $\times (10^{-4})$
w/o TopoLoss	136.0	55.1	40.5	0.218	4.29
with TopoLoss	118.5	55.9	40.1	0.211	3.86

designed latent topologically-regularized diffusion models could boost the accuracy a lot with a slightly lower speed.

Qualitative Visualization: In Fig. 4, we visualize several generated LiDAR point cloud samples from our method TopoLiDM. The visualizations demonstrate the model’s ability to produce realistic and topologically consistent LiDAR scenes, effectively capturing the topological structures of urban driving environments.

C. Conditional LiDAR Generation

Additionally, we explored the potential capability of TopoLiDM to generate LiDAR point clouds conditioned by text descriptions. The visualization in Fig. 5 shows that our model can produce realistic LiDAR scenes that align with the given text inputs, further highlighting its potential for creating training data for downstream fields.

D. Ablation Studies

To analyze contributions of different components in TopoLiDM, we conduct ablation studies on KITTI-360.

Topological Encoder in VAE: We analyze the impact of using a topology-aware encoder versus a standard CNN encoder [17] without topological constraints. The results are presented in Table III. The model with the topology-aware encoder significantly outperforms the model without it, achieving lower FRID and MMD scores. This demonstrates the effectiveness of incorporating topological constraints in the encoder to preserve global topological consistency.

Depth of GNN Layers: We also investigate the impact of the depth of GNN layers in the encoder. The results are shown

TABLE VI: Ablation studies of the of our latent diffusion model.

Method	FRID ↓	FSVD ↓	FPVD ↓	JSD ↓	MMD ↓ $\times (10^{-4})$
w/o LDM	1991	437.1	261.7	0.351	12.43
with LDM	118.5	55.9	40.1	0.211	3.86

TABLE VII: Ablation studies of the number of sampling steps in our diffusion model.

Steps	FRID ↓	FSVD ↓	FPVD ↓	JSD ↓	MMD ↓ $\times (10^{-4})$
25	133.1	60.7	43.5	0.215	4.29
50	118.5	55.9	40.1	0.211	3.86
100	114.5	54.7	39.8	0.214	3.77
200	113.1	53.7	38.9	0.213	3.78

in Table IV. The proposed encoder with 4 layers achieves the best performance. Reducing the depth to 3 layers results in a noticeable drop in performance, particularly in FRID and FSVD, indicating that the model struggles to capture sufficient global topological context.

Topological Loss Analysis: We also evaluate the impact of the topological loss (TopoLoss). The results are shown in Table V, where the model with the topological loss achieves better performance in terms of FRID and MMD, highlighting the importance of enforcing topological consistency.

Diffusion Models: We also conduct ablation studies for the designed latent diffusion models. As in Table VI, we remove the diffusion models, only utilizing topology-aware VAE for the LiDAR generation, which has much worse performance. We also compare different sampling steps in diffusion models in Fig. VII. More sampling steps may slightly increase the generation accuracy but will incur much longer inference time. Thus, we choose the sampling step as 50 in our diffusion model for the trade-off between accuracy and efficiency.

V. CONCLUSION

We proposed TopoLiDM, a novel framework that integrates graph neural networks with diffusion models under topological regularization, significantly enhancing the fidelity and interpretable ability of LiDAR point cloud generation. Extensive experiments on the KITTI-360 dataset demonstrate the competitive performance of our TopoLiDM. For our future work, we would design more condition signals like trajectory, and also evaluate the proposed algorithm on additional datasets.

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